# Optimising Group Sequential Designs:

# Where Frequentist meets Bayes

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**Improving the Efficiency of Clinical Trials**

**From Methods to Practice**

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# Plan for this talk

I shall consider :

Frequentist and Bayesian designs for a group sequential trial

What we would like to do

What we can do

Convergence of approaches

# Problem formulation

Consider a Phase III clinical trial comparing a new treatment against a control.

We denote the treatment effect by $\theta$.

Examples: $\theta$ could be the difference in mean response or, in a time-to-event study, $\theta$ could be the log hazard ratio.

We wish to decide whether $\theta > 0$, in which case the new treatment is superior.

The trial will have $K$ analyses.

The information for $\theta$ at analysis $k$ will be $\mathcal{I}_k$.

Marginally,

$$\widehat{\theta}_k \sim N(\theta, \mathcal{I}_k^{-1})$$

and the score statistics, $S_k = \widehat{\theta}_k \mathcal{I}_k$, have independent increments.

# Decision theoretic formulation: Berry & Ho (*Bmcs*, 1988)

Prior distribution $\pi(\theta)$.

Possible decisions

$$d_1: \quad \text{Do not pursue drug approval,}$$

$$d_2: \quad \text{Pursue drug approval.}$$

Loss function for taking decision $d_1$ or $d_2$ when the true value of the treatment effect is $\theta$

$$L(\theta, d_1) = 0 \qquad \text{for all } \theta,$$

$$L(\theta, d_2) = \begin{cases} -K\theta & \text{if } \theta > 0, \\ L & \text{if } \theta \leq 0. \end{cases}$$

Sampling cost: 1 per subject in the trial, $N$ in total, say.

Aim: minimise the expected loss

$$\int \pi(\theta) \int f(x \mid \theta) \{L(\theta, d(x)) + N(x)\} \, \mathrm{d}x \, \mathrm{d}\theta$$

# Decision theoretic formulation: Berry & Ho (*Bmcs*, 1988)

Aim: minimise the expected loss

$$\int \pi(\theta) \int f(x \mid \theta) \{L(\theta, d(x)) + N(x)\} \, \mathrm{d}x \, \mathrm{d}\theta$$

Berry & Ho found the optimal stopping rule and decision rule by dynamic programming.

They presented examples with priors

$$\theta \sim N(-1, 2), \quad \theta \sim N(0, 2), \quad \theta \sim N(1, 2),$$

$$K = 5,000 \quad \text{and} \quad L = 2,000,$$

and showed results for designs with 2 and 3 analyses.

Lorden (*Ann. Statistics*, 1976) had applied the numerical optimisation scheme described by Lai (*Ann. Statistics*, 1973) to solve similar problems in a frequentist setting.

# A frequentist problem: Barber & Jennison (*Bmka*, 2002)

Test $H_0$: $\theta \leq 0$ against $\theta > 0$.

Specify type I and type II error rates

$$Pr_{\theta=0}\{\text{Reject } H_0\} \leq \alpha, \quad Pr_{\theta=\delta}\{\text{Reject } H_0\} \geq 1-\beta.$$

So, a fixed sample size test requires information

$$\mathcal{I}_{fix} = \{\Phi^{-1}(1-\alpha) + \Phi^{-1}(1-\beta)\}^2/\delta^2.$$

Aim: in a group sequential test with $K$ analyses, minimise

$$\int f(\theta)\, \mathbb{E}(\mathcal{I}_T)\, \mathrm{d}\theta$$

where $f(\theta)$ is a $N(\delta/2, (\delta/2)^2)$ density and $\mathcal{I}_T$ is the observed information on termination.

Barber & Jennison found optimal stopping and decision rules by dynamic programming.

# A frequentist problem: Barber & Jennison (*Bmka*, 2002)

Minimum possible values of $\int f(\theta) E_\theta(\mathcal{I}_T)\,\mathrm{d}\theta$, where $f(\theta)$ is the density of a $N(\delta, \delta^2/4)$ distribution, for group sequential tests with $K$ equally sized groups, $\mathcal{I}_{max} = R\mathcal{I}_{fix}$, type I error probability $\alpha = 0.025$, power $0.9$ at $\theta = \delta$.

**Minimum values of $\int f(\theta) E_\theta(\mathcal{I}_T)\,\mathrm{d}\theta$, as a percentage of $\mathcal{I}_{fix}$**

| | $R$ | | | | | |
|---|---|---|---|---|---|---|
| $K$ | 1.01 | 1.05 | 1.1 | 1.2 | 1.3 | *Minimum over $R$* |
| 2 | 79.3 | 74.7 | **73.8** | 74.8 | 77.1 | 73.8 at $R$=1.11 |
| 3 | 74.8 | 69.0 | 67.0 | **66.1** | 66.6 | 66.1 at $R$=1.20 |
| 5 | 71.1 | 65.1 | 62.7 | 60.9 | **60.5** | 60.5 at $R$=1.32 |
| 10 | 68.2 | 62.1 | 59.5 | 57.5 | **56.7** | 56.4 at $R$=1.46 |
| 20 | 66.8 | 60.6 | 58.0 | 55.8 | **54.8** | 54.2 at $R$=1.59 |

Recommend: $K = 5$, $R = 1.05$ or $1.1$.

# A frequentist problem: Barber & Jennison (*Bmka*, 2002)

For practical application:

Error spending tests with type I and type II error spending functions of the form

$$f(\mathcal{I}) = \alpha\,(\mathcal{I}/\mathcal{I}_{\max})^{\rho}, \quad g(\mathcal{I}) = \beta\,(\mathcal{I}/\mathcal{I}_{\max})^{\rho}$$

are almost optimal.

Error spending designs adapt to observed information levels, controlling the type I error rate and maintaining efficiency.

Rho-family designs with $\rho = 2$ have a sample size "inflation factor" around $R = 1.1$.

Designs with $\rho = 3$ have an "inflation factor" around $R = 1.05$.

# Derivation of optimal tests

The problem is to minimise

$$\int f(\theta)\, E_\theta(\mathcal{I}_T)\, \mathrm{d}\theta,$$

subject to

$$Pr_{\theta=0}\{\text{Reject } H_0\} \leq \alpha, \quad Pr_{\theta=\delta}\{\text{Reject } H_0\} \geq 1-\beta. \qquad (1)$$

Following the Lagrangian approach, we use dynamic programming to minimise

$$\int f(\theta)\, E_\theta(\mathcal{I}_T)\, \mathrm{d}\theta + \lambda_1\, Pr_{\theta=0}\{\text{Reject } H_0\} + \lambda_2\, Pr_{\theta=\delta}\{\text{Accept } H_0\}.$$

Then we search for values $\lambda_1$ and $\lambda_2$ so that (1) is satisfied — and we have the solution to the problem with error rate constraints.

# Derivation of optimal tests

In minimising

$$\int f(\theta)\, E_\theta(\mathcal{I}_T)\, \mathrm{d}\theta + \lambda_1\, Pr_{\theta=0}\{\text{Reject } H_0\} + \lambda_2\, Pr_{\theta=\delta}\{\text{Accept } H_0\},$$

we have solved a Bayes decision problem with prior

$$\begin{array}{ll} \theta = 0 & \text{with probability } 1/3 \\ \theta = \delta & \text{with probability } 1/3 \\ \theta \sim N(\delta/2, \delta^2/4) & \text{with probability } 1/3 \end{array}$$

and a cost of sampling $\mathcal{I}_T$ when $\theta \sim N(\delta/2, \delta^2/4)$.

This is a Bayes optimal procedure — but for a rather odd looking problem.

# Back to Bayesian proposals

Spiegelhalter, Freedman & Parmar (1994) proposed the following sequential procedure. Given a prior $\pi(\theta)$,

At each analysis $k = 1, \ldots, K-1$

| | |
|---|---|
| if $Pr\{\theta > 0 \mid \textit{data}\} > 1 - \epsilon$ | stop, declare $\theta > 0$ |
| if $Pr\{\theta < 0 \mid \textit{data}\} > 1 - \epsilon$ | stop, declare $\theta \leq 0$ |
| otherwise | continue to group $k+1$, |

after group $K$

| | |
|---|---|
| if $Pr\{\theta > 0 \mid \textit{data}\} > 1 - \epsilon$ | stop, declare $\theta > 0$ |
| otherwise | stop, do not declare $\theta > 0$. |

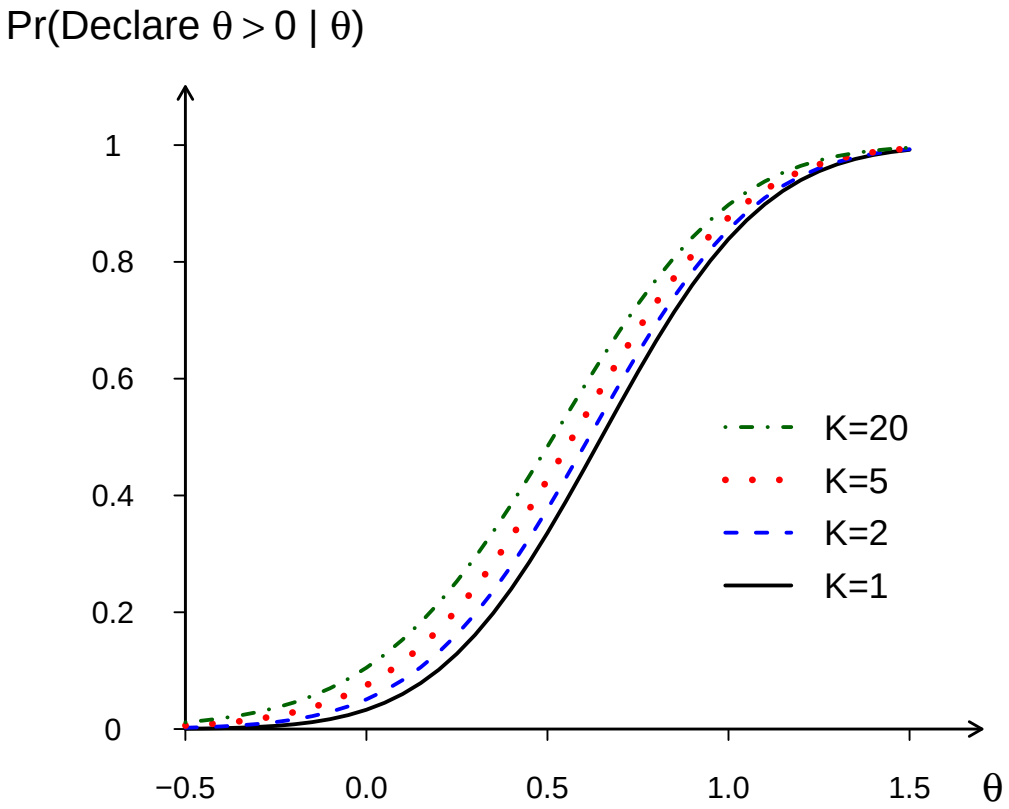
The trial stops early if the $1 - 2\epsilon$ credible interval for $\theta$ does not contain zero.

The credible interval is not affected by the fact that earlier analyses have been conducted.

# Bayesian proposals: Spiegelhalter et al. (1994)

However, the frequentist properties of the Bayes procedure **are** affected by the number of analyses conducted.

Probability of declaring $\theta > 0$ for procedures with prior $\theta \sim N(0.5, 0.5)$, $\epsilon = 0.025$, and 1, 2, 5 and 20 analyses:



# Bayesian proposals: Calibrated procedures

Spiegelhalter et al. (1994) proposed use of a "handicap prior", chosen so that the procedure has a particular type I error rate.

If the number of analyses is $K$ and maximum information is $\mathcal{I}_K$, the handicap prior is

$$\theta \sim N(0, (h\mathcal{I}_K)^{-1}).$$

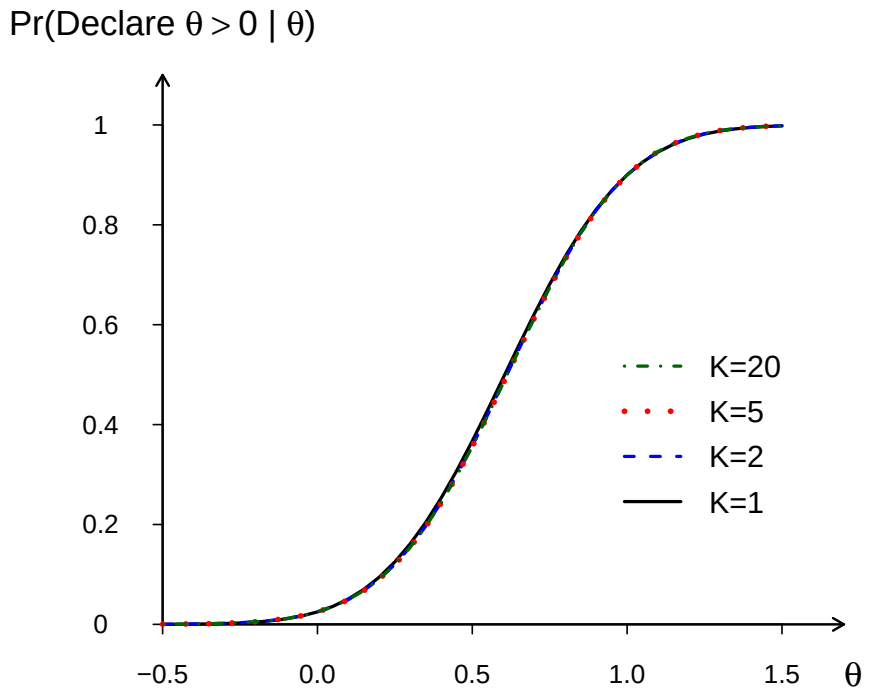
The "handicap" $h$ depends on the number of analyses. Here are values for $\alpha = 0.025$.

| *Number of analyses* $K$ | *Handicap* $h$ |
|---|---|
| 1 | 0.000 |
| 2 | 0.163 |
| 5 | 0.271 |
| 20 | 0.382 |

# Bayesian proposals: Calibrated procedures

We can choose $\mathcal{I}_K$ so that the procedure with a handicap prior has a specific power if the treatment effect is $\theta = \delta$.

Power functions of designs with $Pr\{\text{Reject } H_0 \,|\, \theta = 1\} = 0.9$




Note: When power curves are matched at $\theta = 0$ and $\theta = 1$, they are just about indistinguishable everywhere.

# Bayesian proposals: Ventz & Trippa (2015)

Ventz & Trippa proposed optimising and calibrating at the same time. Their problem formulation has:

Possible decisions

$$d_1: \quad \text{Do not pursue drug approval,}$$
$$d_2: \quad \text{Pursue drug approval.}$$

A "gain function" or "utility" comprising

$$G(\theta, d_1) = 0 \qquad \text{for all } \theta,$$
$$G(\theta, d_2) = \begin{cases} K(\theta) & \text{if } \theta > 0, \\ -L(\theta) & \text{if } \theta \leq 0, \end{cases}$$

plus a term

$$B(\theta, d_i, T)$$

denoting the additional benefit from reaching decision $d_i$ at analysis $T$ minus the cost of treating patients in the trial.

# Bayesian proposals: V&T Calibrated Bayes procedure

Assuming a prior distribution $\pi(\theta)$, Ventz & Trippa seek to maximise the expected gain

$$-\int_{-\infty}^{0} L(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta + \int_{0}^{\infty} K(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta + \int_{-\infty}^{\infty} E\{B(\theta, d_i, T)\}\, \pi(\theta)\, d\theta.$$

subject to error rate constraints

$$Pr(D = d_2 \,|\, \theta = 0) = \alpha \quad \text{and} \quad Pr(D = d_1 \,|\, \theta = \delta) = \beta$$

for a specified value of $\delta$.

Note that type I and type II errors feature twice, in different guises.

# Bayesian proposals: V&T Calibrated Bayes procedure

On Slide 14, we saw an example of how, to a high degree of accuracy, power belong to a two parameter family.

Thus, the constraints

$$Pr(D = d_2 \,|\, \theta = 0) \;=\; \alpha \quad \text{and} \quad Pr(D = d_1 \,|\, \theta = \delta) \;=\; \beta$$

are essentially equivalent to

$$\int_{-\infty}^{0} L(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta \;=\; p_1$$

and

$$\int_{0}^{\infty} K(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta \;=\; p_2$$

for certain values of $p_1$ and $p_2$.

# Bayesian proposals: V&T Calibrated Bayes procedure

Thus, when Ventz & Trippa maximise

$$-\int_{-\infty}^{0} L(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta \;+\; \int_{0}^{\infty} K(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta$$

$$+\; \int_{-\infty}^{\infty} E\{B(\theta, d_i, T)\}\, \pi(\theta)\, d\theta$$

subject to

$$Pr(D = d_2 \,|\, \theta = 0) \;=\; \alpha \quad \text{and} \quad Pr(D = d_1 \,|\, \theta = \delta) \;=\; \beta.$$

they are effectively maximising

$$\int_{-\infty}^{\infty} E\{B(\theta, d_i, T)\}\, \pi(\theta)\, d\theta.$$

subject to

$$\int_{-\infty}^{0} L(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta = p_1 \;\text{ and }\; \int_{0}^{\infty} K(\theta)\, Pr(D = d_2 \,|\, \theta)\, \pi(\theta)\, d\theta = p_2.$$

# Bayesian proposals: V&T Calibrated Bayes procedure

Given the equivalence of the two types of constraint on the power function, they are also maximising

$$\int_{-\infty}^{\infty} E\{B(\theta, d_i, T)\}\, \pi(\theta)\, d\theta.$$

subject to

$$Pr(D = d_2 \,|\, \theta = 0) \;=\; \alpha \quad \text{and} \quad Pr(D = d_1 \,|\, \theta = \delta) \;=\; \beta.$$

This is exactly the type of problem that Jennison and others have tackled from a frequentist perspective.

# A remark on the frequentist designs we saw earlier

I noted that the designs produced by Barber & Jennison were Bayes procedures for rather strange looking priors.

However, we can replace the constraint

$$Pr(D = d_2 \,|\, \theta = 0) \;=\; \alpha \quad \text{and} \quad Pr(D = d_1 \,|\, \theta = \delta) \;=\; \beta$$

by an (almost) equivalent constraint of the form

$$\int_{-\infty}^{0} Pr(D = d_2 \,|\, \theta)\, f(\theta)\, d\theta = p_1 \;\text{ and }\; \int_{0}^{\infty} Pr(D = d_2 \,|\, \theta)\, f(\theta)\, d\theta = p_2,$$

where $f(\theta)$ is a $N(\delta/2,\, (\delta/2)^2)$ density.

Then, we will obtain essentially the same optimal design by solving a more reasonable looking Bayesian problem.

**CONVERGENCE!**

# Conclusions

**Theoretical underpinnings**

The convergence between frequentist and calibrated Bayes procedures demonstrates the "complete class theorems" of Brown, Cohen & Strawderman (*Ann. Statistics*, 1980) which state

$$\{\text{The set of admissible frequentist procedures}\}$$

$$= \{\text{The set of Bayes optimal procedures}\}.$$

**Practical consequences**

Both schools can learn from each other — we are (or should be) solving the same problems.